Architectural Execution Profile: Classical CPU Benchmarks vs. Physical NISQ QPU Bounds

Metric / Architectural Domain	Classical CPU Execution Benchmark	NISQ Quantum Hardware Target (Transmon QPU)
State Space Encoding	$\mathcal{O}(D)$ Explicit Floating-Point Array	$\mathcal{O}(\log_2 D)$ **Hilbert Basis Amplitudes**
Active Working Memory	~16 KB - 302 KB Serialized Cache	**6 Superconducting Transmon Qubits (Wires)**
Single-Query Inference Latency	0.05 ms (SVM) - 12.86 ms (Random Forest)	**CPU Sim: 5.62 ms → QPU Hardware: ~2.5 µs Pulse**
Microwave Gate Durations	N/A (Von Neumann Arithmetic ALU)	**Single-Qubit: 35 ns, CNOT Entangler: 260 ns**
Generalization Gap (Overfit)	+6.9% (SVM) up to +25.1% (Random Forest)	**+4.76% (Unitary Isometry Rotation Bounds)**
Storage Space Compression	Baseline Reference (1× to 270× footprint)	**665× Smaller Footprint than Random Forest (0.45 KB)**